

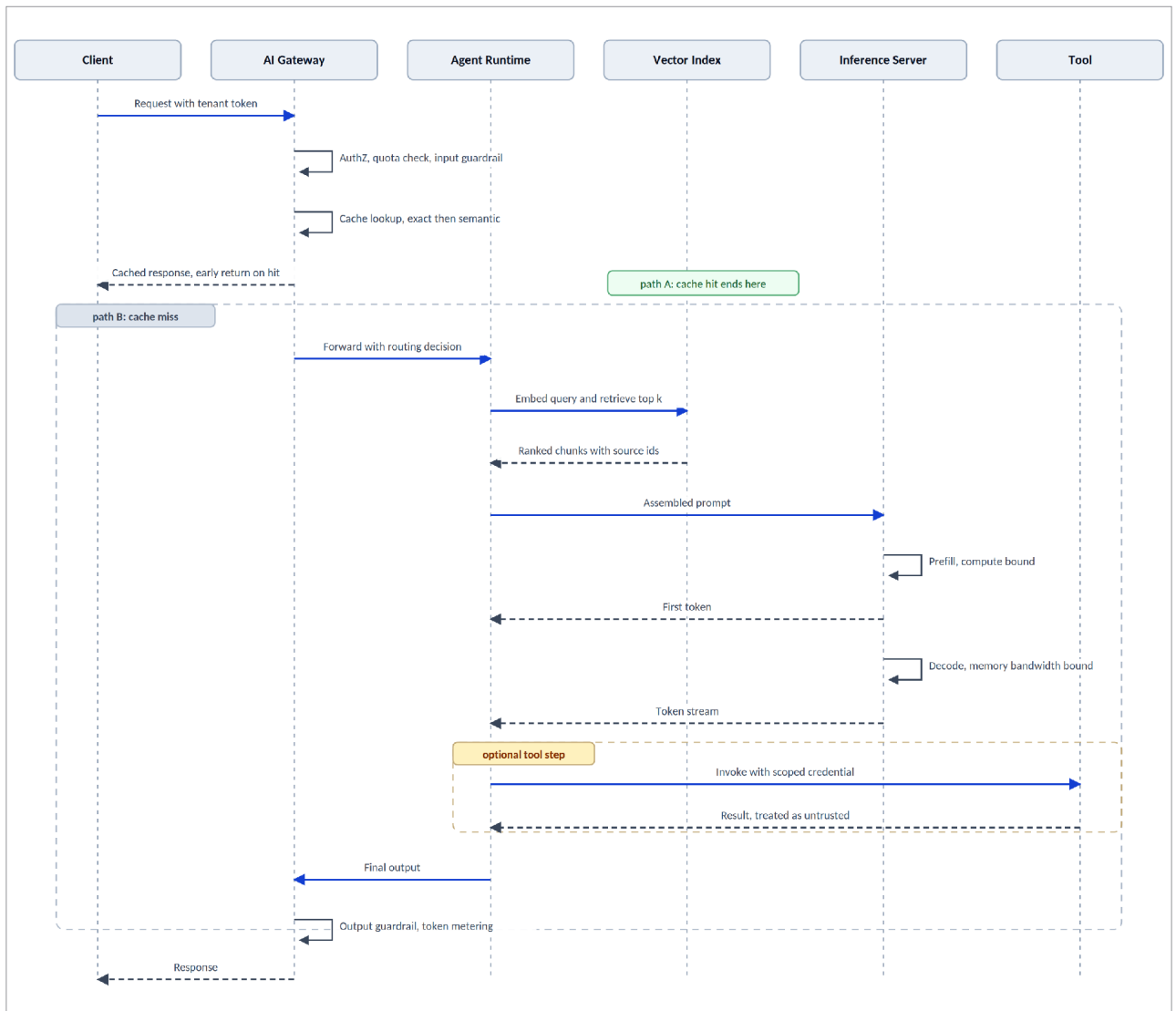


Figure 2: Request lifecycle. Note that the gateway touches the request three times, and that tool output re-enters the model as new untrusted input

- Prompt templates and the tool schemas the model is allowed to call.
- The embedding model, the chunking rules, and the index built from both.
- Guardrail policies and the evaluation datasets that define what passing means.

The embedding model bites hardest. Vectors from one model are not comparable to vectors from another, so swapping it invalidates the entire index. If rebuilding ten million chunks takes six hours, that belongs in a runbook long before someone proposes the upgrade in sprint planning.

Common advice says to treat prompts as code. That

is half right. A prompt without an evaluation set is untested code, and untested code in a non-deterministic system is closer to a configuration guess than a change you can defend in a postmortem.

Hardware scheduling has caught up. Dynamic Resource Allocation reached general availability in Kubernetes 1.34 (released August 2025) and is on by default from 1.35, letting workloads declare the device attributes they need instead of an opaque GPU count. One thing to resist: self-service on accelerator quota.

How do you observe a probabilistic system?

Three audiences need three views, and a single dashboard serving all of them will serve none.